

Progress Report

- Increment 3 -

Group #2

1) Team Members

Colin Maloney – CPM22H – colinpmaloney
Zachary Kupersmith – Zak22 – kupersmithzac
Alex Nino – an23k – alexnino45
Kobus VanSteenburg – kmv21a – kobusthegreat
Samuel Pauly – stp21 – sampauly

2) Project Title and Description

Track Tracker. A data platform that detects emerging music tracks before they hit mainstream charts by monitoring Spotify playlists and Soundcloud trending data.

3) Accomplishments and overall project status during this increment

The frontend has been connected to the database on the backend and successfully displays data retrieved from comparing songs on tiktok videos and spotify songs to display the top songs used in certain hashtags on tiktok. The project accomplishes the goals that we set out to complete at the beginning of this project.

4) Challenges, changes in the plan and scope of the project and things that went wrong during this increment

Some things that were challenging were setting up the docker and correctly retrieving the data from the database from the frontend. The frontend team believed that album covers were being fetched along with the songs, but in reality, the album covers were not being retrieved.

5) Team Member Contribution for this increment

Kobus VanSteenburg contributed to the frontend design and implementation. Kobus worked with Samuel to create all of the main pages of the frontend and implemented the album cover patch for the database. Also created a backend patch to better sort songs for the top charts. Kobus also completed the progress report.

Samuel Pauly contributed to the frontend design and implementation. Samuel worked with Kobus to create all of the main pages of the frontend and implemented the album cover patch for the database.

Colin finished up spotify to database. Added snapshots for individual track stats on every ingestion. Updated to use streams with web scraping instead of popularity score because it is depreciated.

Zack verified DB setup and research/ testing into composite index on tracksnapshot before realizing the lack of necessity and difficulty. Also did Progress report video.

Alex assisted with the ingestion process to capture per-track stat snapshots on each run. Also assisted in research, gathering information for data processing, providing support to help keep things moving.

6) Stakeholder Communication

Dear Stakeholders,

I hope this message finds you well. I am writing to provide an update on our project's progress and outline the next steps in our development timeline. We are pleased to share that the team has successfully completed all major components of the application. This includes the core frontend and backend systems, full integration between these layers, and the implementation of key features that enable users to interact with the platform and access real-time data. Additionally, external data sources have been fully integrated, allowing the application to deliver dynamic and relevant information as intended.

Throughout development, we encountered and addressed challenges related to third-party service integration and data consistency across components. By refining our data handling processes and strengthening communication between system elements, we have ensured a stable and cohesive architecture.

Looking ahead, our focus will shift toward final testing, performance optimization, and user interface refinements to ensure a smooth, intuitive, and polished user experience. These efforts will prepare the application for a successful release. At this stage, the project remains on track and is well-positioned for completion. We appreciate your continued support and look forward to sharing the finalized product with you soon.

Please feel free to reach out if you have any questions or would like further details.

Sincerely,
Team Track Tracker

7) Link to video

<https://youtu.be/6CHXUysmPII>